Phase 16: Formal Business Rules Catalog

Document Code: BRULE-CAT-001

Project: Duplicate Bin Ecosystem

Standard: ISO/IEC 25010 / Business Rules Group (BRG) Standard

Format: Rule ID | Rule Statement | Business Rationale | Source Code Reference | Affected Module | Exception

1. Business Rules Catalog

BRULE-001: Master Preservation Invariant

- **Rule Statement:** Within any duplicate cluster containing N identical files ($N \ge 2$), exactly one file must be designated as the master file (`keep=True`) and excluded from deletion. Under no circumstances may all N files be selected for archiving simultaneously.
- **Business Rationale:** Prevents total data loss. If all copies of a duplicate group were deleted, the user would lose the asset entirely.
- **Source:** duplicate_bin/core/selection.py#L35-L65, duplicate_bin/ui/main_window.py#L320
- **Affected Module:** Selection Engine & Results Tree UI.
- **Exception:** None. System UI explicitly disables checking the final box in a group.

BRULE-002: Operating System Root Protection

- **Rule Statement:** The system shall reject scanning or file modification requests directed at operating system root directories and critical system filesystems, specifically:
- Windows: `C:\Windows`, `C:\Program Files`, `C:\Program Files (x86)`, `C:\ProgramData`.
- Linux/POSIX: `/bin`, `/sbin`, `/usr`, `/lib`, `/etc`, `/dev`, `/proc`, `/sys`.
- macOS: `/System`, `/Library`, `/Applications`.
- **Business Rationale:** Prevents users or rogue scripts from corrupting system DLLs, kernel modules, or executable binaries that share identical sizes.
- **Source:** duplicate_bin/core/utils.py#L22-L48, docs/TECHNICAL_ARCHITECTURE.md#L142
- **Affected Module:** Pre-Scan Directory Validator (`is_safe_path`).

- **Exception:** Explicit user whitelisting in developer debug mode (`--danger-allow-system-paths`), requiring terminal warning confirmation.

BRULE-003: Pairing Code Expiration Window

- **Rule Statement:** A 6-digit numeric device pairing code generated by the SaaS web portal shall remain valid for exactly 300 seconds (5 minutes) from its creation timestamp.
- **Business Rationale:** Limits the exposure window against brute-force guessing and unauthorized workstation claiming.
- **Source:** backend/app/main.py#L82, backend/app/main.py#L93-L96
- **Affected Module:** SaaS Device Pairing Controller.
- **Exception:** None. Once 300 seconds elapse, code is purged from the in-memory dictionary.

BRULE-004: Deduplication Credit Debit Lifecycle

- **Rule Statement:** Deduplication credits (`credit_bytes`) shall only be deducted from the user's account ledger upon the initial transition of a scan record to `status == 'completed'`. In-progress scans, failed scans, and repeated queries on already completed scans must not debit credits.
- **Business Rationale:** Protects customers from double billing and ensures credits are only consumed when measurable value (reclaimable duplicate storage) is delivered.
- **Source:** backend/app/main.py#L163-L170, backend/app/main.py#L195-L197
- **Affected Module:** SaaS Scan Telemetry & Billing Ledger.
- **Exception:** Administrative credit reconciliation adjustments performed by support staff.

BRULE-005: Quarantine Sandbox Boundary Enforcement

- **Rule Statement:** File movement during archiving operations must be confined to the designated application sandbox directory (`~/.duplicate_bin/archive/`). Canonical target paths must be verified using `os.path.realpath` to guarantee they do not escape the sandbox boundary via directory traversal or symlink redirection.

- **Business Rationale:** Prevents malicious symlink attacks or arbitrary file overwrite vulnerabilities.
- **Source:** docs/TECHNICAL_ARCHITECTURE.md#L141, duplicate_bin/engine/archive.py#L42
- **Affected Module:** Safe Archive Engine.
- **Exception:** None. Violations raise an immediate `SecurityError` and halt execution.

BRULE-006: Regular Expression Complexity Guardrail

- **Rule Statement:** User-supplied regular expression patterns in the file filtering dialog must not exceed 512 characters in length.
- **Business Rationale:** Mitigates Regular Expression Denial of Service (ReDoS) attacks caused by catastrophically backtracking regex patterns.
- **Source:** docs/TECHNICAL_ARCHITECTURE.md#L144, duplicate_bin/core/filtering.py#L38
- **Affected Module:** File Filter Engine.
- **Exception:** None. Inputs exceeding 512 characters are truncated and flagged with an input error.

BRULE-007: Image Quality Selection Precedence

- **Rule Statement:** When the `KEEP_BEST_QUALITY` strategy is selected, file extensions classified as uncompressed Camera RAW formats (`.cr2`, `.nef`, `.arw`, `.dng`) shall always take precedence over lossless formats (`.tiff`, `.png`), which in turn take precedence over lossy compressed formats (`.jpg`, `.jpeg`, `.webp`), regardless of file creation timestamps.
- **Business Rationale:** Guarantees that photographers retain original high-dynamic-range source negatives while purging generated web exports.
- **Source:** duplicate_bin/core/selection.py#L48-L72
- **Affected Module:** Selection Strategy Engine.
- **Exception:** If all files in a group share identical formats, selection falls back to the largest file size or newest modification timestamp.

BRULE-008: Cache Invalidation Compound Key

- **Rule Statement:** A cached cryptographic hash is valid if and only if the triple compound key `(file_path, file_size, mtime)` exactly matches the current operating system filesystem attributes. If `st_mtime` or `st_size` diverges, the cache entry is marked stale and the file must be re-hashed.
- **Business Rationale:** Prevents stale cache reads from erroneously classifying modified files as duplicates.
- **Source:** duplicate_bin/engine/cache.py#L45-L70, docs/TECHNICAL_ARCHITECTURE.md#L98
- **Affected Module:** Persistent SQLite Cache.
- **Exception:** None.